



International School

Capstone Project 1

CMU-SE 450

C1SE.12

Project Proposal

Version 1.1

Date: August 29th, 2026

FashionFit AI – AI-Powered 3D Anthropometric Fitting System

Submitted by

Thang, Le Van	29219042914
Quan, Hoang Ho Dinh	29211125668
Hoa, Nguyen Khac	29219046990
Quynh, Nguyen Truc	29204142567
Tai, Thai Anh	29211162810

Approved by

Thai, Le Minh, Ph.D.

Proposal Review Panel Representative:

Name	Signature	Date
------	-----------	------

Capstone Project 1 - Mentor:

Name	Signature	Date
------	-----------	------

Da Nang, 08/2026

PROJECT INFORMATION			
Project acronym	FFAI		
Project Title	FashionFit AI – AI-Powered 3D Anthropometric Fitting System		
Start Date	August 26, 2026	End Date	Dec 12, 2026
Lead Institution	International School, DuyTan University		
Project Mentor	Thai, Le Minh, Ph.D. Email: vietintel.vn@gmail.com Tel: 0916.544.455		
Scrum master / Project Leader & contact details	Thang, Le Van Email: levanthang0166@gmail.com ID: 29219042914 Tel: 0333792162		
Partner Organization	Duy Tan University		
Project Web URL			
Team members			
ID	Name	Email	Tel
29211125668	Quan, Hoang Ho Dinh	hoangquan9688@gmail.com	0383317427
29219046990	Hoa, Nguyen Khac	hoa590472@gmail.com	0345093728
29204142567	Quynh, Nguyen Truc	trucquynh300905@gmail.com	0392804829
29211162810	Tai, Thai Anh	11a6thaiantai@gmail.com	0815178225

DOCUMENT APPROVALS

The following signatures are required for approval of this document.

Thang, Le Van Student ID: 29219042914 <i>Scrum Master</i>	Signature	Date
Hoa, Nguyen Khac Student ID: 29219046990 <i>Team Member</i>	Signature	Date
Quan, Hoang Ho Dinh Student ID: 29211125668 <i>Team Member</i>	Signature	Date
Quynh, Nguyen Truc Student ID: 29204142567 <i>Team Member</i>	Signature	Date
Tai, Thai Anh Student ID: 29211162810 <i>Team Member</i>	Signature	Date

REVISION HISTORY

Version	Date	Comments	Author
1.0	August 25th, 2026	Initial Draft	FashionFit AI Team
1.1	August 29th, 2026	First Proposal Submission	FashionFit AI Team

Contents

1. Project Title.....	8
2. Team Member.....	8
3. Supervisor(s)	9
4. Problem Statement	9
4.1. What challenges are users/customers facing?.....	9
4.2. Impacts of the problem	9
4.3. Why did the research team choose to address this problem?	10
5. Survey / Existing Solutions	10
5.1. Modern Sizing & Measurement Platforms	10
5.2. FashionFit AI Solution.....	11
5.3. Future Development Direction	11
6. Objectives and Scope.....	11
6.1. Objectives	11
6.2. Scope and Exclusion.....	12
7. Key Feature & Requirements	13
8. Constraints	14
9. Target Users / Stakeholders.....	15
9.1. Online Shopper / Customer.....	15
9.2. E-Commerce Fashion Retailer / Brand.....	15
9.3. Bespoke Tailor / Custom Apparel Maker	15
9.4. Fashion Business Manager / Analyst.....	15
9.5. System Admin.....	15
10. Technology Stack.....	16
11. Methodology & Development Plan	16
12. System Architecture Overview.....	18
13. Potential Risks and Mitigation Strategies	19
14. Expected Outcomes / Deliverables	20
15. References	20
Academic Papers & Standards:.....	20
Scrum & Engineering Practices:	21
Technical Frameworks & Libraries:.....	21

List of Figures

Figure 1. System Context Diagram for FashionFit AI Platform	18
--	----

List of Tables

Table 1. Membership information and roles.....	8
Table 2. Information of Supervisor.....	9
Table 3. Key Features	13
Table 4. Project Constraints.....	14
Table 5. Schedule.....	17
Table 6. Risk of project.....	19

1. Project Title

FashionFit AI – AI-Powered 3D Anthropometric Fitting System

2. Team Member

Table 1. Membership information and roles

TEAM MEMBER	Information & Detailed Responsibilities
Thang Le Van	Email: levanthang0166@gmail.com Student ID: 29219042914 Role: Scrum Master / Project Leader / Fullstack Developer Description: Directs project milestones, coordinates Scrum activities, architects system workflow, develops fullstack modules, and oversees database integration and end-to-end testing.
Quan Hoang Ho Dinh	Email: hoangquan9688@gmail.com Student ID: 29211125668 Role: AI & Computer Vision Engineer Description: Implements Nasion landmark detection, OpenCV reference scaling, and body volume constraints.
Hoa Nguyen Khac	Email: hoa590472@gmail.com Student ID: 29219046990 Role: 3D Graphics & Frontend Developer Description: Builds interactive Three.js 3D canvas, controls parametric mesh morphing, and develops React components.
Quynh Nguyen Truc	Email: trucquynh300905@gmail.com Student ID: 29204142567 Role: UI/UX Designer / QA & QC Tester Description: Designs Figma UI wireframes, crafts pose guidance overlays, and conducts usability testing.
Tai Thai Anh	Email: 11a6thaianhantai@gmail.com Student ID: 29211162810 Role: Backend Developer / DevOps Engineer Description: Builds RESTful API services, manages Supabase database, and configures Docker deployment.

3. Supervisor(s)

Table 2. Information of Supervisor

Full Name: Thai, Le Minh	Academic Degree: Ph.D.
Faculty: International School, Duy Tan University	
Contact Address: Room 204, 2nd Floor – Building G, 120 Hoang Minh Thao Street, Duy Tan University	
Phone: 0916.544.455	Email: vietintel.vn@gmail.com

4. Problem Statement

4.1. What challenges are users/customers facing?

In current online fashion retail and custom tailoring, consumers encounter four major technical and practical hurdles:

- **Lack of accurate sizing verification:** Sizing standards vary drastically across apparel brands (e.g., Size M in Asia vs. Size M in Europe), making manual size selection unreliable.
- **Manual measurement errors (> 5.0 cm):** Untrained users measuring themselves with physical tape measures regularly place measuring bands at incorrect anatomical landmarks or apply uneven tension.
- **The "Puffy Hair" optical distortion (2.0 – 5.0 cm error):** Standard computer vision systems compute body height from the top of the bounding box (vertex), which is severely distorted by voluminous hairstyles, buns, curls, or hats.
- **Clothing occlusion (The Loose/Tight paradox):** Traditional 2D segmentation models only detect outer garment contours, failing to distinguish between a slender person wearing loose clothes and a larger person wearing tight clothes.

4.2. Impacts of the problem

- **For online shoppers:** Causes frustration from receiving ill-fitting garments, leading to time-consuming return processes and reduced confidence in digital fashion.

- **For fashion retailers:** Sizing discrepancies cause 30% – 40% return rates, generating immense reverse logistics expenses, stock devaluation, and margin degradation.

- **For bespoke tailors:** Requires mandatory in-person appointments, limiting geographic customer reach and increasing operational overhead.

- **For the environment:** Unnecessary transportation of returned garments and discarded fast-fashion clothing significantly elevate carbon emissions and textile waste.

4.3. Why did the research team choose to address this problem?

- **Global trend:** Applying Computer Vision and 3D digital human anthropometry to e-commerce is rapidly accelerating worldwide to automate sizing and digital fittings.

- **Feasibility:** The project combines accessible client-side pose tracking (MediaPipe) with mathematical anthropometry and Three.js 3D mesh rendering, providing high precision without expensive dedicated hardware.

- **Commercial and social value:** Solving sizing uncertainty directly reduces e-commerce return waste, empowers local fashion businesses, and democratizes bespoke tailoring access.

5. Survey / Existing Solutions

5.1. Modern Sizing & Measurement Platforms

- **Manual Sizing Tables:** Basic measurement charts on e-commerce websites. They rely entirely on self-measured data and fail to provide personalized recommendations.
- **3D Laser Body Scanner Booths:** High-end physical hardware booths that produce sub-millimeter accurate body meshes. However, they cost over \$15,000 per unit and require physical customer attendance.
- **Basic 2D Camera Measurement Apps:** Smartphone apps using simple edge detection. They suffer heavily from background noise, hair volume distortion, and cannot accurately calculate 3D circumferences.

5.2. FashionFit AI Solution

FashionFit AI introduces an intelligent, dual-image anthropometric measurement framework:

- **Facial Nasion Bone Calibration:** Leverages a hybrid MediaPipe Pose and Face Mesh pipeline to accurately detect cranial Nasion/Orbit landmarks, establishing an immutable skeletal height anchor and eliminating hair volume distortions.
- **Volumetric Mathematical Constraints:** Employs biological body density ($\rho \approx 1.01 \text{ g/cm}^3$) along with user Weight and Gender to deduce true body girths beneath clothing.
- **ISO 8559 Cross-Section Slicing:** Extracts circumference curves at standardized anatomical planes (Chest, Waist, Hips).
- **Interactive 3D Mannequin Canvas:** Projects calculated body metrics onto a customizable 3D avatar in real time using Three.js.

5.3. Future Development Direction

- **Augmented Reality (AR) Virtual Try-On:** Integrating 3D cloth physics simulation to drape garments directly over the custom body mesh.
- **E-Commerce Widget Plugin:** Packaging the system into lightweight Shopify/WooCommerce extensions for instant merchant integration.
- **B2B Industrial Pattern Export:** Exporting standardized CAD patterns directly to textile manufacturing lines.

6. Objectives and Scope

6.1. Objectives

The system provides automated, non-invasive body measurement extraction and size recommendation to optimize online garment purchases.

- **Predict and extract accurate measurements:** Automatically compute 7+ body parameters (Height, Shoulder Width, Arm Length, Inseam/Outseam, Chest, Waist, Hips) with **< 2.0 cm** circumference error.

- **Eliminate physical distortions:** Apply facial bone anchoring and volume constraints to overcome puffy hair and clothing occlusion.
- **Visualize interactive 3D human form:** Render an anatomically synchronized 3D avatar within < 3.0 seconds on web browsers.
- **Ensure technical and academic viability:** Deliver a fully tested web platform with Dockerized services within the August – December 2026 academic timeline.

6.2. Scope and Exclusion

6.2.1. In-Scope

- **End-Users (Shoppers):** Capture dual photos with live MediaPipe guidance, view extracted measurements, interact with 3D avatar, and receive recommended clothing sizes.
- **Tailors & Fashion Retailers:** Access customer measurement dossiers, manage standard garment size tables, and export ISO 8559 dimension reports.
- **System Administrators:** Manage accounts, monitor processing latency, and oversee cloud database storage.

6.2.2. Out-of-Scope

- Real-time dynamic cloth collision and fabric elasticity simulation.
- Production of physical hardware scanning booths or dedicated depth sensors.
- Direct financial payment gateway processing and logistics shipping management.

7. Key Feature & Requirements

Table 3. *Key Features*

No	Name	Description
1	User Authentication & Profile	Supports secure email/password and Google OAuth2 login. Manages personal measurement histories and biometric preferences.
2	Live Pose Alignment Guidance	Employs client-side MediaPipe Pose to provide visual bounding guides, ensuring 0° front and 90° side angles before capture.
3	Reference Object & Metric Calibration	Detects standard reference markers (A4 paper / ID card) via OpenCV to compute exact pixel-to-millimeter scale factors.
4	Nasion & Volume-Constrained Extraction	Anchors height at skull landmarks and applies weight/gender volumetric penalties to extract ISO 8559 girth measurements. Incorporates a dynamic BMI sanity check ($14 \leq \text{BMI} \leq 45$) and an explicit user disclaimer modal to mitigate manual weight entry discrepancies before executing volumetric penalty calculations.
5	Interactive 3D Mannequin Canvas	Dynamically morphs male/female 3D mesh avatars in Three.js based on extracted metrics with 360° rotation and zoom controls.
6	Garment Size Matching & Reporting	Compares extracted body metrics with merchant size charts (S/M/L/XL) and generates downloadable measurement dossiers.

Quality Attribute Outline: The system ensures high usability (intuitive capture guide), fast performance (< 3s extraction), and strong data privacy (AES-256 encryption).

8. Constraints

Table 4. *Project Constraints*

Constraints	Constraints Description	Guidelines for Acceptance
Economic	Requires budget for cloud hosting, database storage, and testing reference materials. Must remain accessible without expensive camera equipment.	Utilize free academic tiers and cost-effective cloud services (Supabase, Vercel). Keep total operational costs within the approved capstone budget.
Environment	Consumes electrical energy during client processing and server compute. Indirectly decreases carbon footprint by reducing product return shipping.	Optimize frontend JavaScript execution and use efficient OpenCV/MediaPipe pipelines to avoid unnecessary server computation.
Ethical	Processes sensitive full-body user photographs and biometric parameters. Requires strict user consent and transparent algorithmic processing.	Apply data minimization, obtain explicit user consent, enforce SSL/TLS encryption, and allow users to delete stored photos.
Public Health, Safety and Welfare	Body measurement systems can impact user self-image and mental well-being. System must prevent biased or derogatory body classification.	Present sizing data objectively as garment-fitting recommendations without body-shaming terminology. Ensure UI accessibility standards.
Social and Global	Aims to reduce global sizing mismatch, lower textile return waste, and support seamless cross-border fashion commerce.	Ensure responsive design across low-end and high-end mobile devices, supporting standard international measurement units (cm/inch).
Cultural	Must respect Vietnamese and international cultural privacy preferences regarding personal body image capture.	Provide clear instructions allowing users to wear everyday fitted clothing during image capture without compromising privacy.
Sustainability	The software codebase and 3D rendering pipeline must be maintainable for future feature extensions and model updates.	Follow clean code standards, modular React/TypeScript architecture, Dockerized environments, and comprehensive Git documentation.

9. Target Users / Stakeholders

9.1. Online Shopper / Customer

- **Description:** Consumers purchasing apparel across online e-commerce platforms.

- **Benefit:** Quickly discovers exact personal body measurements and selects the perfect garment size without physical try-ons.

9.2. E-Commerce Fashion Retailer / Brand

- **Description:** Online clothing stores, fashion brands, and marketplace merchants.

- **Benefit:** Slashes return logistics costs by up to 60%, boosts customer checkout confidence, and increases sales conversion.

9.3. Bespoke Tailor / Custom Apparel Maker

- **Description:** Tailoring businesses and custom clothing designers.

- **Benefit:** Obtains accurate client measurement profiles remotely, cutting down manual consultation time and broadening client reach.

9.4. Fashion Business Manager / Analyst

- **Description:** Retail decision-makers overseeing inventory and sizing optimization.

- **Benefit:** Reviews aggregated sizing distribution data to optimize manufacturing production quantities.

9.5. System Admin

- **Description:** Technical staff maintaining software operations and cloud infrastructure.

- **Benefit:** Manages access controls, oversees database integrity, and monitors system performance metrics.

10. Technology Stack

The project utilizes the following modern technology stack organized by functional layers:

- **Frontend:** React 18, TypeScript, Tailwind CSS, Lucide React.
- **3D Graphics Engine:** Three.js, WebGL, custom rigged male/female base meshes (.glb).
- **Computer Vision & AI:** MediaPipe Pose (33 body landmarks), MediaPipe Face Mesh (crop-based 468 landmarks for high-precision Nasion cranial anchoring), OpenCV.js (reference contour scaling), One Euro Filter (jitter reduction).
- **Backend Services:** Node.js (Express) API Gateway and Python mathematical modules.
- **Database & Storage:** Supabase (PostgreSQL with Row Level Security) and Supabase Secure Storage.
- **Cloud & DevOps:** Docker, Docker Compose, Vercel, GitHub Actions.
- **Management & Design Tools:** Jira / Trello (Agile board), Figma (UI/UX design), Postman (API testing).

11. Methodology & Development Plan

• **Development Model:** The project follows the **Agile Scrum** framework across 6 two-week sprints. Each sprint encompasses backlog grooming, sprint planning, development, testing, and retrospective review.

• **Sprint Plan (Timeline):** The 16-week execution plan is structured into specific development milestones:

- To mitigate algorithmic bottlenecks in volumetric derivation (Sprint 3) and 3D mesh morphing (Sprint 4), a 4-day contingency research buffer is scheduled, while 3D base mesh rigging runs in parallel starting from Sprint 1.

Table 5. Schedule

No	Task	Duration	Starting Day	Ending Day
1	Initial	10 days	20 Aug 2026	29 Aug 2026
1.1	Requirement Gathering & Literature Review	4 days	20 Aug 2026	23 Aug 2026
1.2	Proposal Document & Feasibility Analysis	6 days	24 Aug 2026	29 Aug 2026
2	Start-up	10 days	30 Aug 2026	08 Sep 2026
2.1	Kick-off Meeting & Environment Setup	2 days	30 Aug 2026	31 Aug 2026
2.2	Software Requirements Specification (SRS) & UI Wireframes	8 days	01 Sep 2026	08 Sep 2026
3	Development (6 Sprints)	84 days	09 Sep 2026	01 Dec 2026
3.1	Sprint 1: Frontend Scaffold & OpenCV Reference Scaling	14 days	09 Sep 2026	22 Sep 2026
3.2	Sprint 2: MediaPipe Pose Alignment & Nasion Landmark Pipeline	14 days	23 Sep 2026	06 Oct 2026
3.3	Sprint 3: Volumetric Penalty Algorithm & Metric Extraction	14 days	07 Oct 2026	20 Oct 2026
3.4	Sprint 4: Three.js 3D Mannequin Canvas & Mesh Morphing	14 days	21 Oct 2026	03 Nov 2026
3.5	Sprint 5: Garment Sizing Engine & Supabase Integration	14 days	04 Nov 2026	17 Nov 2026
3.6	Sprint 6: End-to-End System Integration & ISO 8559 Evaluation	14 days	18 Nov 2026	01 Dec 2026
4	Project Retrospective & Testing	7 days	02 Dec 2026	08 Dec 2026
5	Deployment & Final Defense Presentation	4 days	09 Dec 2026	12 Dec 2026

12. System Architecture Overview

FashionFit AI connects Online Shoppers, Tailors/Retailers, and System Administrators into a unified anthropometric measurement ecosystem. The user captures dual-view photographs under real-time MediaPipe pose validation. The system extracts metric coordinates using OpenCV reference scaling and Nasion height calibration, enforces volumetric penalties via user weight and gender, and extracts standardized ISO 8559 circumferences. Results are rendered on an interactive Three.js 3D avatar and saved to the Supabase database.

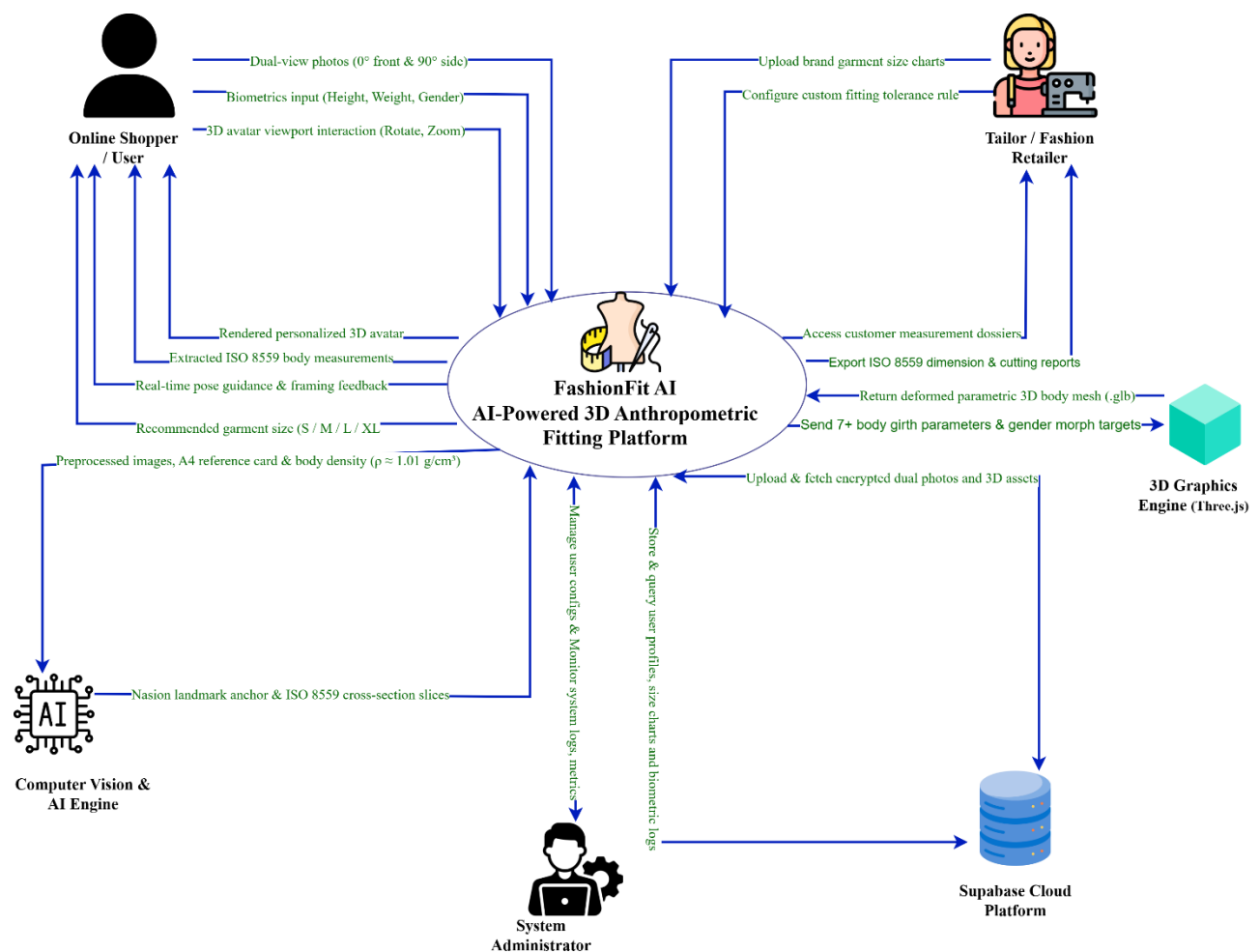


Figure 1. System Context Diagram for FashionFit AI Platform

13. Potential Risks and Mitigation Strategies

Table 6. Risk of project

Risk	Definition	Level	Likelihood	Mitigation Strategy
Data & Lighting Risk	Poor ambient lighting and cluttered backgrounds cause landmark detection failure.	Medium	High	Apply adaptive histogram equalization and Bilateral Filtering in OpenCV to sharpen body contours.
Pose Alignment Risk	Users stand at incorrect angles or distances from the camera.	High	Medium	Enforce real-time MediaPipe skeletal angle checks; disable capture button until user matches the guide overlay.
Model Accuracy Risk	Clothing folds introduce noise in waist and chest circumference estimation.	High	Medium	Incorporate weight/gender volumetric mathematical constraints to deduce true contours beneath fabrics.
Performance Risk	Intensive 3D mesh rendering causes browser stuttering on low-end mobile devices.	Medium	Medium	Optimize 3D base meshes to low-poly format and use requestAnimationFrame throttling in Three.js.
Security & Privacy Risk	User body photographs could be intercepted or exposed.	High	Low	Enforce client-side in-memory processing; photos are evaluated strictly in browser RAM and immediately garbage-collected without server transmission. If transient storage is opted, an automated Supabase database hook triggers permanent deletion within 60 seconds.
Technical Capability Risk	Team members learning 3D WebGL morphing and anthropometric formulas may encounter delays.	High	Medium	Conduct weekly knowledge-sharing sessions, follow modular component design, and consult academic supervisor regularly.

14. Expected Outcomes / Deliverables

- **Web Application:** A fully functional, responsive web application supporting dual-image capture, automated anthropometric extraction, and interactive 3D mannequin visualization.
- **Source Code:** Modular and fully documented codebase hosted on GitHub with complete Docker Compose deployment files.
- **Technical Documentation & Demo Video:** Comprehensive engineering documentation (SRS, SDD, Test Suite) alongside an end-to-end video demonstration.
- **Final Report:** Detailed Capstone 1 graduation report covering algorithmic methodology, ISO 8559 measurement accuracy evaluations, and future roadmap.

15. References

Academic Papers & Standards:

- [1] [ISO] ISO 8559-1:2017, *"Size designation of clothes — Part 1: Anthropometric definitions for body measurement,"* International Organization for Standardization, Geneva, Switzerland, 2017.
- [2] [arXiv] C. Lugaresi et al., *"MediaPipe: A Framework for Building Perception Pipelines,"* arXiv preprint arXiv:1906.08172, 2019.
- [3] [IEEE] G. Pavlakos et al., *"Expressive Body Capture: 3D Hands, Face, and Body from a Single Image,"* in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 10975–10985.
- [4] [ACM] G. Casiez, N. Roussel, and D. Vogel, *"I € Filter: A Simple Speed-based Low-pass Filter for Noisy Input in Interactive Systems,"* in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '12), 2012, pp. 2527–2530.
- [5] [IEEE] IEEE Standard for Software and System Test Documentation, *IEEE Std 829-2008*, 2008.

Scrum & Engineering Practices:

- [6] Ken Schwaber and Jeff Sutherland, *"The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game,"* Scrum.org, 2020.
- [7] Atlassian, *"What is Scrum?"* Atlassian Agile Coach, 2024.

Technical Frameworks & Libraries:

- [8] Three.js Authors, *"Three.js JavaScript 3D Library Documentation,"* Threejs.org, 2025.
- [9] React Team, *"React — A JavaScript library for building user interfaces,"* React.dev, 2025.
- [10] Supabase Inc., *"Supabase Documentation: The Open Source Firebase Alternative,"* Supabase.com, 2025.